

OJT PROJECT DOCUMENTATION

Project Overview: SteelSnap [Centervert]

Steel Snap is a web application designed to autonomously generate a bill of materials by identifying steel beams, columns, and joists from PDF structural plans. Within this larger application architecture, our team is specifically assigned to engineer the core backend machine learning and computer vision pipeline that drives the application's detection capabilities.

Technical Stack for Interns

- **Languages:** Python
- **Computer Vision & ML:** YOLO-family models (ONNX format), LightGBM, scikit-learn
- **Data Processing & Geometry:** Pandas, NumPy, PyArrow, SciPy, Shapely
- **Document Processing:** PyMuPDF
- **Tools & Version Control:** GitHub, Codex CLI

Core Pipeline Responsibilities

1. Data Engineering & Ground Truth Generation

- **Structured Annotation:** Reviewing complex structural PDF drawings to accurately identify and mark relevant structural elements.
- **Data Export:** Exporting verified structural annotations into JSON format to serve as structured ground-truth data for evaluating the Steel Snap detection engine.

2. Detection Engine

- **Applied Machine Learning:** Utilizing a combination of geometric detection, raster/vector PDF analysis, and computer vision (YOLO models via ONNX Runtime) to localize structural features like beam lines and column markers.
- **Feature Engineering:** Applying thresholding, calibration, and candidate scoring (via LightGBM) to refine the pipeline's detection accuracy before it reaches the web client.
- **Algorithmic Testing:** Comparing geometric, CV-based, and ML-based detection approaches to determine the most robust method for handling varying PDF drawing conditions.

3. Benchmarking & Model Evaluation

- **Automated Workflows:** Running specialized Python scripts (benchmark.py, export_ml_candidates.py, diff_candidate_runs.py) to systematically compare detection outputs and measure model behavior across different configurations.

- **Performance Metrics:** Recording and interpreting critical data science metrics, including precision, recall, F1 score, and overall accuracy, to validate model improvements before deployment.

4. Advanced Error Analysis

- **Behavioral Audits:** Moving beyond simple binary validation by analyzing how the engine reacts to real-world architectural variables (e.g., beam/column layouts, line weights, similar-looking symbols, and rasterized vs. vector-based pages).
- **Failure Mode Identification:** Investigating the root causes of false positives, false negatives, duplicate detections, and inaccurate bounding boxes by correlating model outputs with specific drawing conventions, label placements, and element densities.
- **Code-to-Output Validation:** Reviewing technical configurations and codebase changes via private GitHub repositories to map specific threshold adjustments or model updates to their resulting behavior in the pipeline's outputs.

Sample of Data:

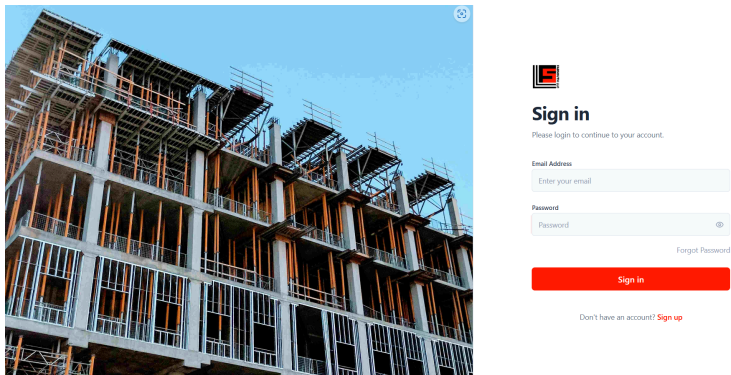
S_SET_Results.json

```

7      "y1": 342.72916412353516,
8      "x2": 1489.2009662370626,
9      "y2": 331.5717899719116
10     },
11     "label": "HSS12X8X1/2",
12     "member_type": "beam",
13     "structural_type": "hss",
14     "source": "auto"
15   },
16   {
17     "source_pdf": "S.SET.pdf",
18     "page": 10,
19     "coords": {
20       "x1": 1488.6183098057957,
21       "y1": 331.61790271155644,
22       "x2": 1496.6666870117188,
23       "y2": 415.7291717529297
24     },
25     "label": "HSS12X8X1/2",
26     "member_type": "beam",
27     "structural_type": "hss",
28     "source": "manual"
29   },
30   {
31     "source_pdf": "S.SET.pdf",
32     "page": 10,
33     "coords": {
34       "x1": 1473.6666870117188,
35       "y1": 413.2291717529297,
36       "x2": 1496.5718679163358,
37       "y2": 414.7382447327416
38     },
39     "label": "HSS12X8X1/2",
40     "member_type": "beam",
41     "structural_type": "hss",
42     "source": "manual"
43   },

```

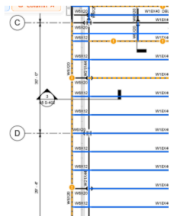
Sample of Web app (still in development)



Sample of short documentation of use-cases

Issue: Labeled lines are not detected and beams have incorrect length.

Screenshot:



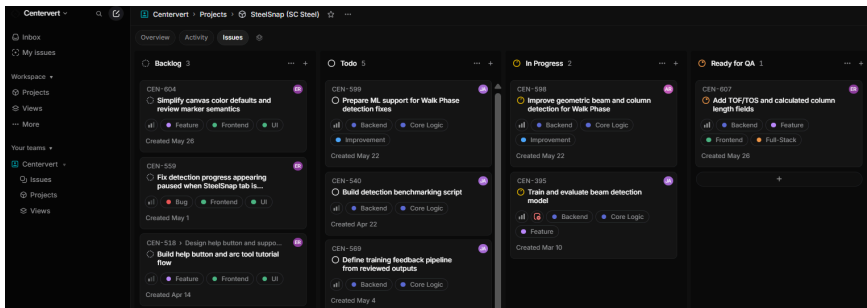
Sharepoint link:

<https://centervert-my.sharepoint.com/my?viewid=d8c89819%2D718d%2D44c4%2D4434%2D44953b7cab&id=%2Fpersonal%2Fpharold%2Fcentervert%2Fcom%2FDocuments%2FDETECTION%20TESTING%2FStructural%20%2D%20CUP%2FMISS%20ED%20BEAMS%2Fpng&parent=%2Fpersonal%2Fpharold%2Fcentervert%2Fcom%2FDocuments%2FDETECTION%20TESTING%2FStructural%20%2D%20CUP>

Short Explanation:

Some lines with labels are not detected because the arrows interfere with the detection process. Some beams also have incorrect lengths because the system mistakenly identifies the gridlines as the beam endpoints.

Tickets via Linear:



[Codes/modules are made private]